Academic year: 2026-2027

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ROADMAP – Project 3 MANET/IoT routing in ns-3: AODV vs OLSR vs DSDV under density and mobility

Variant: Standard Core ns-3 Implementation

1. Team Responsabilities

GIUROESCU Cristian-Andrei: **NS-3 Configuration & Scenario Automation**

- Building the primary ns-3 scenario script, configuring the IEEE 802.11g PHY/MAC layer (ConstantRateWifiManager, 6 Mbps), and implementing the Random Waypoint mobility models. The midterm goal is to ensure the parameterization (protocol, node count, speed, seed) works correctly from the command line.

GORAN Alexandru-Mihai: **Data Logging & KPI Extraction**

- Integrating FlowMonitor to capture metrics, converting FlowMonitor XML outputs to CSV, and writing Python scripts to compute PDR, delay, and jitter. The midterm goal is to produce the baseline delay and PDR plots for the 50-node static scenario.

DIACONU Andrei: **Batch Execution & Routing Overhead**

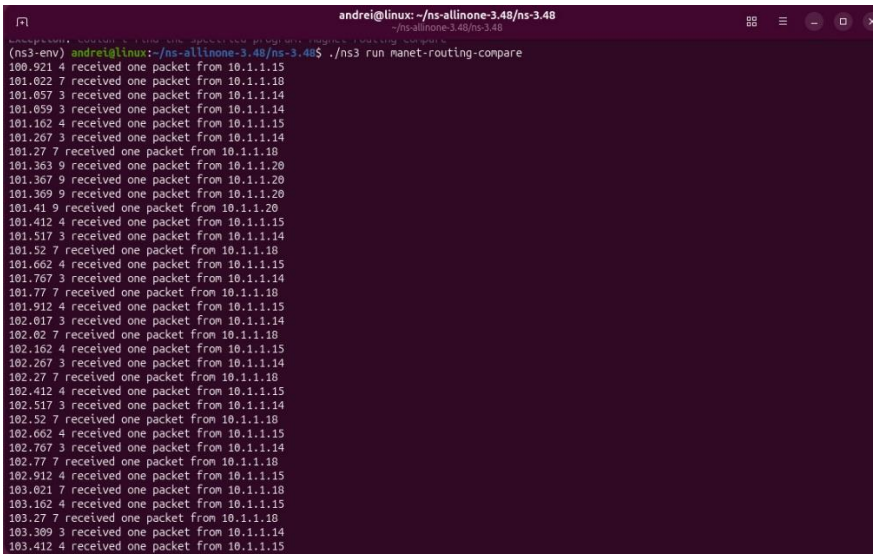
- Developing the batch script to run the 75-run matrix with parallel seeds, and extracting routing control packets from ns-3 traces or promiscuous sniffer callbacks to calculate the normalized routing load. The midterm goal is to complete the automated pipeline that accepts CSV files and outputs the final plotted results.

2. P0 Evidence: Foundations

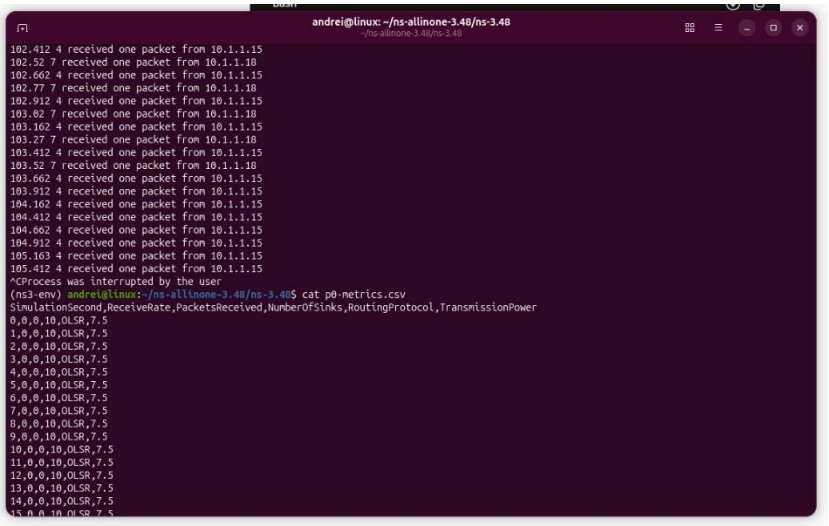
Our group has successfully built ns-3 and verified its core functionality by executing and understanding the base MANET routing example.

- ./ns3 run manet-routing-compare -> This runs the default MANET routing example built into ns-3. By executing this, we verified that the simulator compiles correctly and can successfully simulate

a wireless ad-hoc network scenario. The terminal output showed a continuous stream of packet transmissions between different node IP addresses.



- ./ns3 run "manet-routing-compare --protocol=OLSR --flowMonitor=true --CSVfileName=p0-metrics.csv"; cat p0-metrics.csv



We reviewed things like SimulationSecond, ReceiveRate & PacketsReceived, NumberOfSinks, RoutingProtocol and TransmissionPower.